

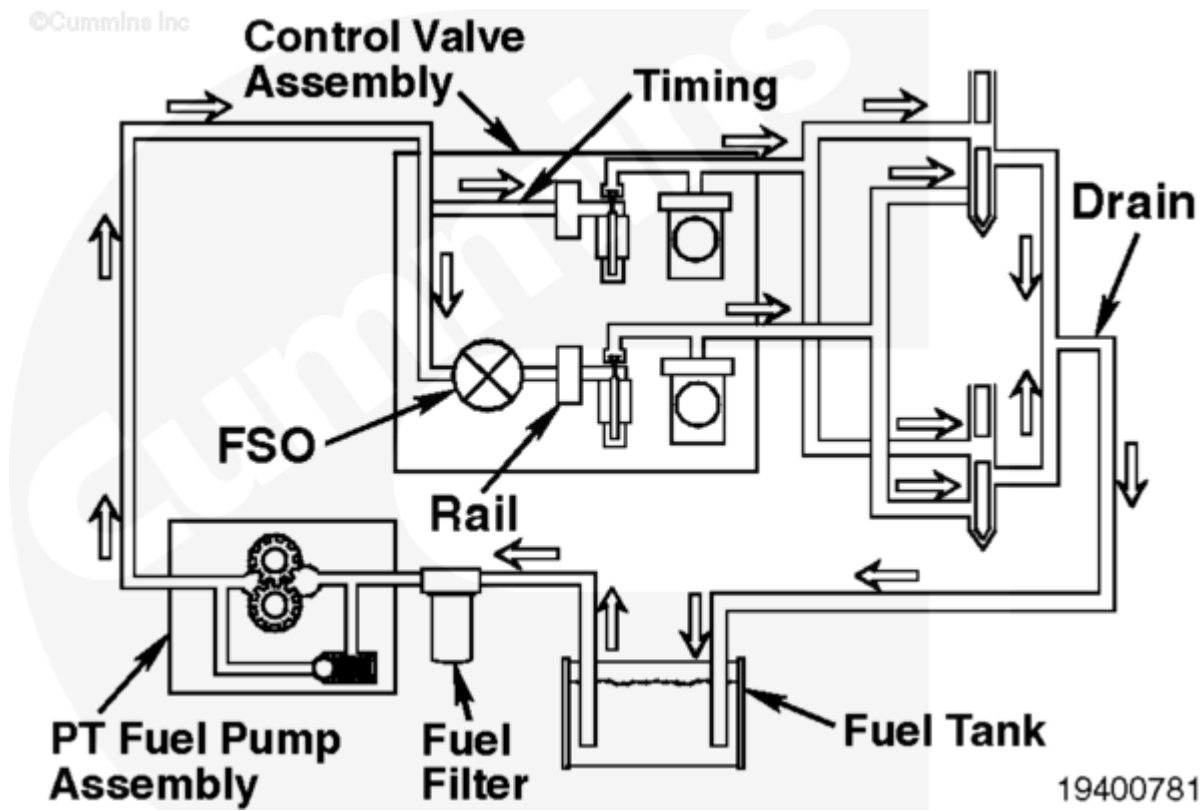
Fault Code: 514

Fuel Rail Actuator - Mechanically Stuck

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 514 PID(P): SPN: FMI: Lamp: Shutdown SRT:	Fuel control valve - mechanically stuck. This was formerly called the fuel rail actuator circuit. The error between the estimated rail fueling and desired rail fueling is outside of acceptable limits.	Engine will overspeed, run at one speed, or will not run.



Quantum Fuel System Flow Schematic

Circuit Description

The ECM uses the fuel rail pressure signal and the engine speed to estimate the actual fueling that the engine is receiving, and then constantly compares this value to the desired fueling for the given speed and load. When there is too large of an error in these values for too long, then Fault Code 514 is logged.

Component Location

The fuel rail actuator is located on the left-side (bottom) of the ECVA.

Shoptalk

The estimated fuel rail and the desired fuel rail parameters can be monitored on the electronic service tool. This fault is a check on the ECM's control of the rail actuator and subsequent fuel flow. If the desired fuel rail can **not** be met by commanding more current to the actuator, or if the desired fuel rail is being exceeded and can **not** be reduced by reducing the current to the actuator, then Fault Code 514 is logged. This fault is **not** logged when:

- Engine speed is below 1200 rpm
- Coolant temperature is below 0°C [32°F]
- Fault Code 451 or 452 is active. Once the Run/Stop switch is cycled, the fault becomes inactive.

Refer to Troubleshooting Fault Code t05-514 (</qs3/pubsys2/xml/en/procedures/01/01-t05-514.html>)